

Bath-heat and action-current fluctuation study (production)

August 27, 2026

Question and scope

We test two distinct questions using the same projection-free Cartesian trajectory. The thermodynamic diagnostic uses the bath heats and medium entropy, while the action current is analyzed as a separate finite-time observable. Correlation between the two does not by itself identify action current with entropy production. The recorded quantities are

$$J_E(t) = \frac{Q_L - Q_R}{2t}, \quad \Sigma_t^m = -\frac{Q_L}{T_L} - \frac{Q_R}{T_R}, \quad J_M(t) = t^{-1} \int_0^t j_M(s) ds.$$

For $T_L = 10$ and $T_R = 2$, the heat-current affinity is $\Delta\beta = T_R^{-1} - T_L^{-1} = 0.4$. At finite time,

$$\frac{\Sigma_t^m}{t} = \Delta\beta J_E(t) - \frac{1}{2}(T_L^{-1} + T_R^{-1}) \frac{\Delta E}{t} + O(\text{integration error}).$$

Executive result

The direct finite-time medium-entropy diagnostic is consistent with its unit-slope reference in 0 of 14 raw-count-qualified fits; the bath-heat diagnostic contains its $\Delta\beta = 0.4$ reference in 0 of 16 qualified fits. These counts are a resolution summary, not a binary theorem test. The scientific interpretation must also use the time trend, the fitted range, and the omitted system-entropy boundary term. Plus-four tail probabilities are used only to display zero or very small counts. All symmetry fits below require nonzero raw counts on both sides.

Simulation and numerical audit

The production sampler uses $c_j = x_j + iy_j$, $\gamma = 0.1$, $dt = 5 \times 10^{-4}$, burn-in $B = 500$, and non-overlapping base blocks of length $t = 20$. There is no action-floor projection. Each block records $Q_L, Q_R, \Delta E, \Sigma_t^m$, the action current, stream ID, and block ID.

n	blocks	$\langle J_M \rangle$	$\langle J_E \rangle$	$\langle \Sigma^m/t \rangle$	balance RMS
10	1,000,064	0.39809	3.21212	1.28485	9.71e-06
20	1,000,064	0.11979	1.40691	0.56275	5.71e-06
30	1,000,064	0.05426	0.79595	0.31839	3.99e-06
40	1,000,064	0.03052	0.52294	0.20918	3.05e-06

Table 1: Production means and discrete first-law residual.

Control runs and cross-checks

Uncertainties in this table are standard errors of independent stream means, so temporal blocks within a stream are not treated as independent replicates. The timestep pairs test discretization; the equal-temperature run tests zero-current symmetry; and swapping the baths must reverse the currents while leaving mean entropy production nonnegative.

control	n	(T_L, T_R)	dt	$J_M \pm \text{SE}$	$J_E \pm \text{SE}$	$\Sigma^m/t \pm \text{SE}$	balance RMS
driven $dt = 5 \times 10^{-4}$	20	(10,2)	5.00e-04	0.1213 $\pm$ 0.0023	1.4205 $\pm$ 0.0278	0.5641 $\pm$ 0.0115	5.88e-06
driven $dt = 2.5 \times 10^{-4}$	20	(10,2)	2.50e-04	0.1213 $\pm$ 0.0027	1.4042 $\pm$ 0.0316	0.5570 $\pm$ 0.0120	1.40e-06
$n = 40$, $dt = 5 \times 10^{-4}$	40	(10,2)	5.00e-04	0.0311 $\pm$ 0.0012	0.5244 $\pm$ 0.0184	0.2087 $\pm$ 0.0076	2.90e-06
$n = 40$, $dt = 2.5 \times 10^{-4}$	40	(10,2)	2.50e-04	0.0289 $\pm$ 0.0010	0.5087 $\pm$ 0.0163	0.2075 $\pm$ 0.0071	7.45e-07
equal temperature	20	(6,6)	5.00e-04	-0.0033 $\pm$ 0.0021	-0.0306 $\pm$ 0.0216	4.34e-04 $\pm$ 0.0016	2.44e-06
baths swapped	20	(2,10)	5.00e-04	-0.1225 $\pm$ 0.0027	-1.4201 $\pm$ 0.0310	0.5683 $\pm$ 0.0131	5.32e-06

Table 2: Projection-free Cartesian validation runs. All midpoint failure counts are zero; halving dt reduces the first-law residual.

The independent canonical action-current implementation gives $J_M = 0.1181 \pm 0.0025$ for $n = 20$, compared with the Cartesian value 0.1213 ± 0.0023 . The difference is 0.94 combined standard errors. The canonical implementation is used only as an action-current cross-check, not for the bath-heat or entropy-production analysis.

Medium-entropy and heat-current symmetry

For the entropy-production rate $a = \Sigma_t^m/t$, the plotted symmetry function is

$$R_t(a) = t^{-1} \log \left[\frac{p_t(a)}{p_t(-a)} \right].$$

The simple asymptotic reference has slope one. For J_E , the corresponding reference slope is $\Delta\beta = 0.4$.

n	t	N	negative	slope	95% CI	R^2	bins
10	20	1,000,064	7,158	0.6529	[0.6378, 0.6675]	0.9956	21
20	20	1,000,064	107,491	0.3408	[0.3364, 0.3446]	0.9973	60
20	40	499,968	10,195	0.4128	[0.3591, 0.4626]	0.7535	60
30	20	1,000,064	214,942	0.2467	[0.2445, 0.2488]	0.9930	60
30	40	499,968	48,561	0.3165	[0.3104, 0.3219]	0.9931	60
40	20	1,000,064	276,375	0.2081	[0.2062, 0.2098]	0.9938	60
40	40	499,968	85,778	0.2541	[0.2504, 0.2574]	0.9957	60
40	80	249,984	15,572	0.3280	[0.3143, 0.3419]	0.9743	60
40	160	124,928	781	0.3292	[-0.0229, 0.6795]	0.8483	4

Table 3: Raw-count-qualified medium-entropy symmetry fits. Missing rows have insufficient two-sided counts.

Fit-range robustness

The predeclared fixed-bin result remains the primary diagnostic. As a robustness check, a second estimator uses symmetric equal-width bins whose range is set by the 99th percentile of the positive and negative-magnitude samples separately, with the number of bins limited by rare-side effective support. Differences between the two estimates diagnose curvature or fit-window sensitivity; the estimator closer to a reference slope is not selected post hoc.

Table 4: Primary fixed-range and adaptive-range robustness estimates.

n	t	observable	fixed slope	robust slope	robust 95% CI	bins
10	20	Σ^m/t	0.6529	0.6501	[0.6304, 0.6670]	35
10	20	J_E	0.2173	0.2191	[0.2128, 0.2247]	37
10	20	J_M	1.2202	1.0368	[1.0221, 1.0483]	45
10	40	J_M	0.9961	1.2852	[1.1100, 1.4770]	3
20	20	Σ^m/t	0.3408	0.3640	[0.3614, 0.3666]	60
20	20	J_E	0.1258	0.1222	[0.1213, 0.1230]	60
20	20	J_M	0.8926	0.7998	[0.7942, 0.8050]	60
20	40	Σ^m/t	0.4128	0.4894	[0.4755, 0.5016]	42
20	40	J_E	0.1561	0.1557	[0.1520, 0.1588]	39
20	40	J_M	1.2483	1.1315	[1.1155, 1.1449]	49
20	80	J_M	0.7174	1.3207	[1.1680, 1.4792]	7
30	20	Σ^m/t	0.2467	0.2600	[0.2581, 0.2618]	60
30	20	J_E	0.0811	0.0793	[0.0787, 0.0798]	60
30	20	J_M	0.6431	0.6089	[0.6042, 0.6134]	60
30	40	Σ^m/t	0.3165	0.3341	[0.3302, 0.3373]	60
30	40	J_E	0.1095	0.1079	[0.1067, 0.1089]	60
30	40	J_M	0.9444	0.8823	[0.8731, 0.8902]	60
30	80	Σ^m/t	–	0.4361	[0.4112, 0.4575]	30
30	80	J_E	0.0852	0.1430	[0.1356, 0.1495]	31
30	80	J_M	1.3375	1.2804	[1.2528, 1.3049]	40

Continued on next page

n	t	observable	fixed slope	robust slope	robust 95% CI	bins
30	120	Σ^m/t	–	0.4799	[0.3809, 0.5821]	3
30	120	J_E	–	0.1569	[0.1359, 0.1784]	5
30	120	J_M	–	1.4126	[1.3090, 1.5193]	16
30	160	J_M	–	1.7952	[1.5615, 2.0402]	3
40	20	Σ^m/t	0.2081	0.2182	[0.2166, 0.2197]	60
40	20	J_E	0.0584	0.0567	[0.0562, 0.0572]	60
40	20	J_M	0.5146	0.4960	[0.4919, 0.5007]	60
40	40	Σ^m/t	0.2541	0.2653	[0.2625, 0.2679]	60
40	40	J_E	0.0805	0.0788	[0.0780, 0.0795]	60
40	40	J_M	0.7362	0.7097	[0.7014, 0.7169]	60
40	80	Σ^m/t	0.3280	0.3390	[0.3308, 0.3462]	47
40	80	J_E	0.1074	0.1074	[0.1053, 0.1092]	48
40	80	J_M	1.0886	1.0572	[1.0402, 1.0711]	51
40	120	Σ^m/t	–	0.4012	[0.3730, 0.4309]	25
40	120	J_E	0.0974	0.1241	[0.1179, 0.1296]	27
40	120	J_M	1.3340	1.3273	[1.2977, 1.3523]	37
40	160	Σ^m/t	0.3292	0.4514	[0.3862, 0.5143]	6
40	160	J_E	0.1405	0.1429	[0.1243, 0.1612]	8
40	160	J_M	1.0924	1.4272	[1.3492, 1.4970]	26
40	200	J_M	–	1.7499	[1.5749, 1.9213]	8

Action-current tails and symmetry

The central 1–99% histogram is compared with a fitted normal density. Each empirical upper and lower survival tail is also compared with the corresponding normal CDF. The action-current symmetry slope is reported empirically, without assigning it the thermodynamic reference value one.

n	t	N	negative	slope	95% CI	bins
10	20	1,000,064	19,815	1.2202	[1.1319, 1.2940]	60
10	40	499,968	509	0.9961	[0.1652, 1.7817]	3
20	20	1,000,064	153,016	0.8926	[0.8857, 0.8991]	60
20	40	499,968	23,622	1.2483	[1.2158, 1.2824]	60
20	80	249,984	1,082	0.7174	[0.2495, 1.1758]	7
30	20	1,000,064	277,240	0.6431	[0.6378, 0.6483]	60
30	40	499,968	78,989	0.9444	[0.9334, 0.9546]	60
30	80	249,984	11,913	1.3375	[1.2834, 1.3845]	60
40	20	1,000,064	347,224	0.5146	[0.5093, 0.5192]	60
40	40	499,968	125,818	0.7362	[0.7273, 0.7447]	60
40	80	249,984	32,159	1.0886	[1.0695, 1.1070]	60
40	120	166,656	10,362	1.3340	[1.2873, 1.3732]	60
40	160	124,928	3,604	1.0924	[-0.1775, 2.4038]	11

Table 5: Raw-count-qualified action-current symmetry fits.

Descriptive Gaussian tail fit

In addition to the Gaussian benchmark fixed by the full-sample mean and variance, one Gaussian is fitted jointly to both raw tails with one-sided probabilities at most 0.01. The shared parameters prevent independent fits from hiding positive/negative-tail asymmetry. Because exceedance probabilities at adjacent thresholds are nested and correlated, the fit is a descriptive shape diagnostic rather than a formal goodness-of-fit test.

n	N	sample μ	sample σ	tail μ	tail σ	RMSE ₊	RMSE ₋	points (+/-)
10	1,000,064	0.3981	0.2101	0.4380	0.2383	0.4379	0.6886	59/64
20	1,000,064	0.1198	0.1243	0.1314	0.1490	0.5251	0.5515	39/39
30	1,000,064	0.0543	0.0955	0.0625	0.1200	0.6181	0.7623	32/34
40	1,000,064	0.0305	0.0792	0.0400	0.1016	0.6193	0.6848	28/28

Table 6: Joint two-tail Gaussian fit for the base $t = 20$ blocks. RMSE is computed on the log-probability scale over the fitted tail window.

Dependence on averaging time and threshold

For each threshold with at least 20 raw tail events, probability at most 0.2, and at least four resolved averaging times, we fit the descriptive large-deviation form

$$\log P[J_t \geq A] = c_+(A) - tI_+(A), \quad \log P[J_t \leq -A] = c_-(A) - tI_-(A).$$

The samples at different t are aggregations of the same base streams, so the fitted R^2 values diagnose linearity rather than supply an independent-sample significance test. Missing rows indicate direct-sampling resolution limits, not zero probability.

n	tail	resolved A range	fits	$R^2 \geq 0.98$	rate range
10	+	[0.46, 0.81]	36	36	[0.004, 0.100]
10	−	−	0	0	−
20	+	[0.15, 0.33]	19	19	[0.004, 0.104]
20	−	[0.01, 0.06]	6	6	[0.067, 0.108]
30	+	[0.08, 0.20]	13	13	[0.008, 0.110]
30	−	[0.01, 0.08]	8	8	[0.040, 0.109]
40	+	[0.05, 0.15]	11	11	[0.008, 0.113]
40	−	[0.01, 0.08]	8	8	[0.029, 0.108]

Table 7: Resolved descriptive fits of $\log P$ versus averaging time.

Heat–action coupling

Tight coupling would require more than a large Pearson correlation: the heat-on-action regression must be stable with t , and the residual variance fraction must become small. The joint plots and tables report all three quantities.

n	t	Pearson r	heat-on-action slope	residual fraction
10	20	0.8087	6.2895	0.3459
10	200	0.9336	7.9691	0.1283
20	20	0.5397	5.0939	0.7087
20	200	0.8960	10.5224	0.1972
30	20	0.2647	2.8357	0.9299
30	200	0.8034	11.1207	0.3546
40	20	0.1006	1.2232	0.9899
40	200	0.6831	10.1999	0.5334

Table 8: Heat–action coupling at the shortest and longest requested windows.

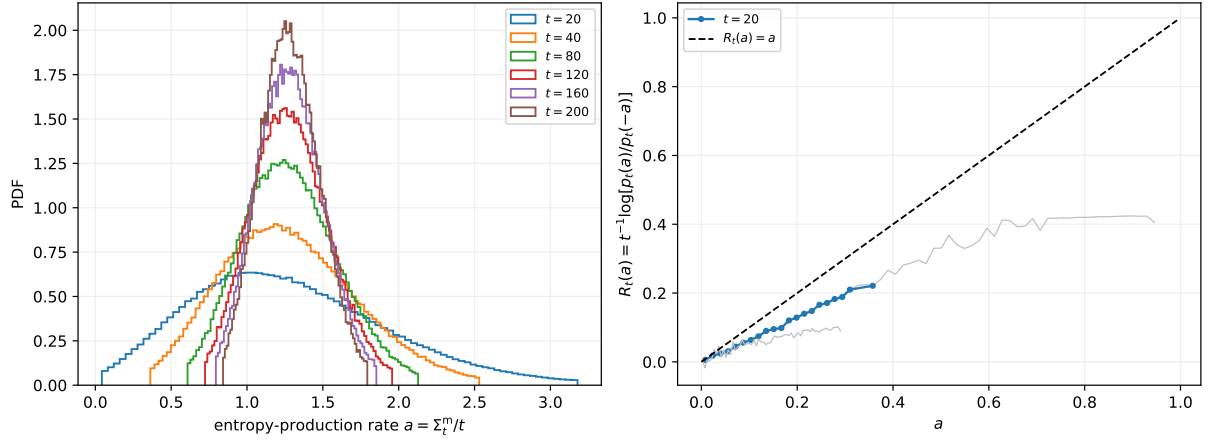


Figure 1: Medium-entropy PDF and raw/plus-four symmetry diagnostic for $n = 10$.

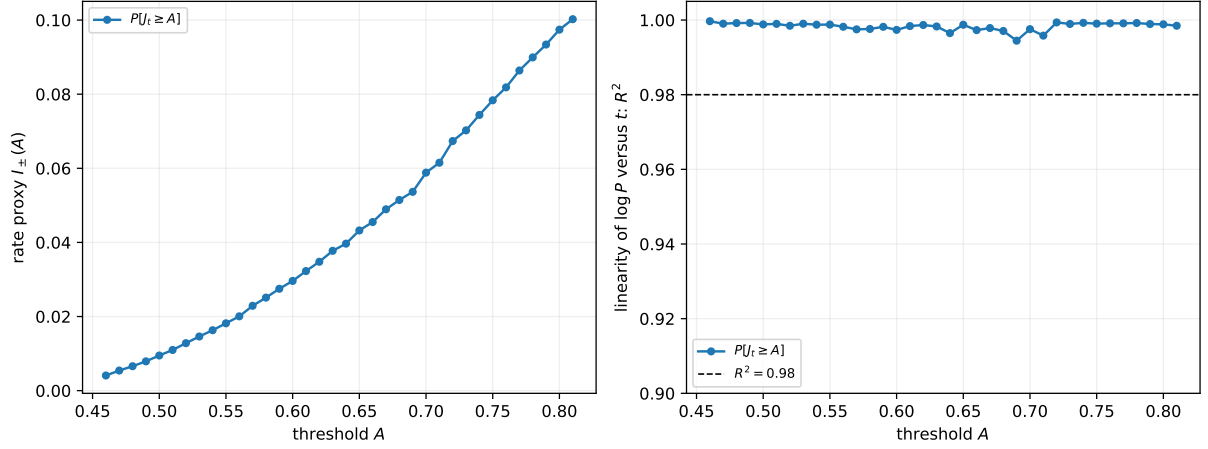


Figure 2: Descriptive action-tail rate proxies and $\log P$ -versus- t linearity for $n = 10$. Only raw-count-qualified rare-tail windows are shown.

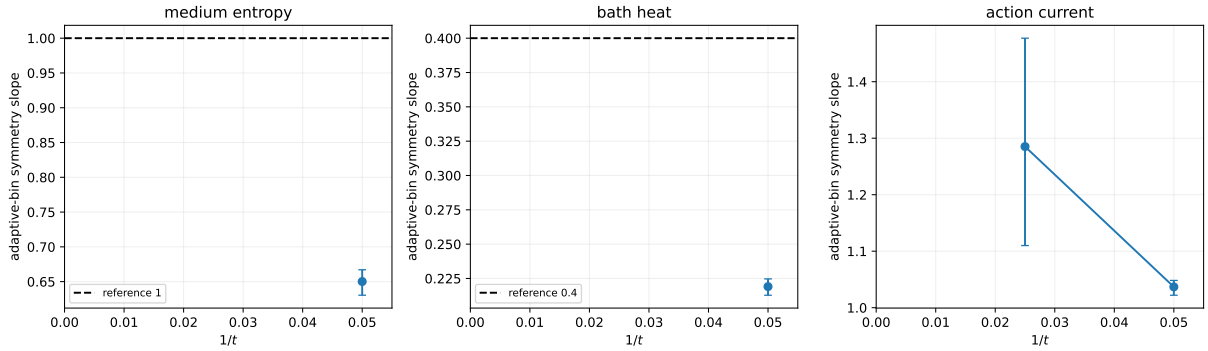


Figure 3: Adaptive-range symmetry-slope robustness check for $n = 10$. The horizontal references apply only to medium entropy and bath heat.

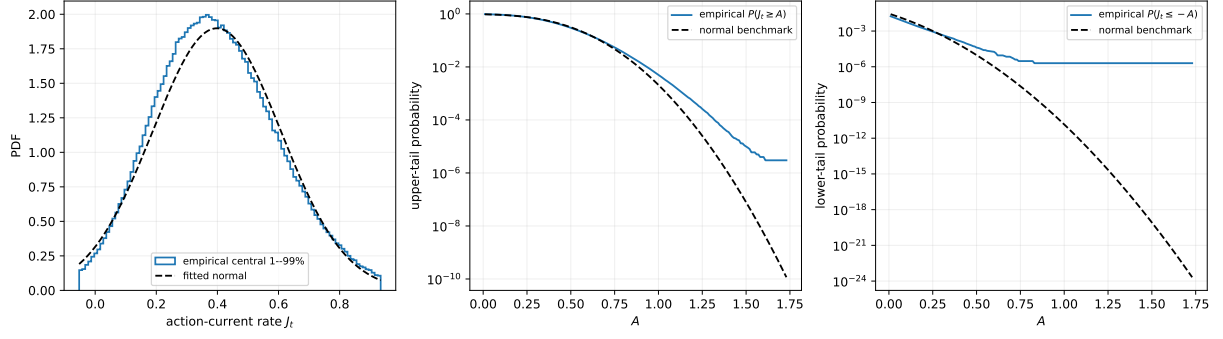


Figure 4: Central action-current histogram and both normal-tail benchmarks for $n = 10$ at $t = 20$.

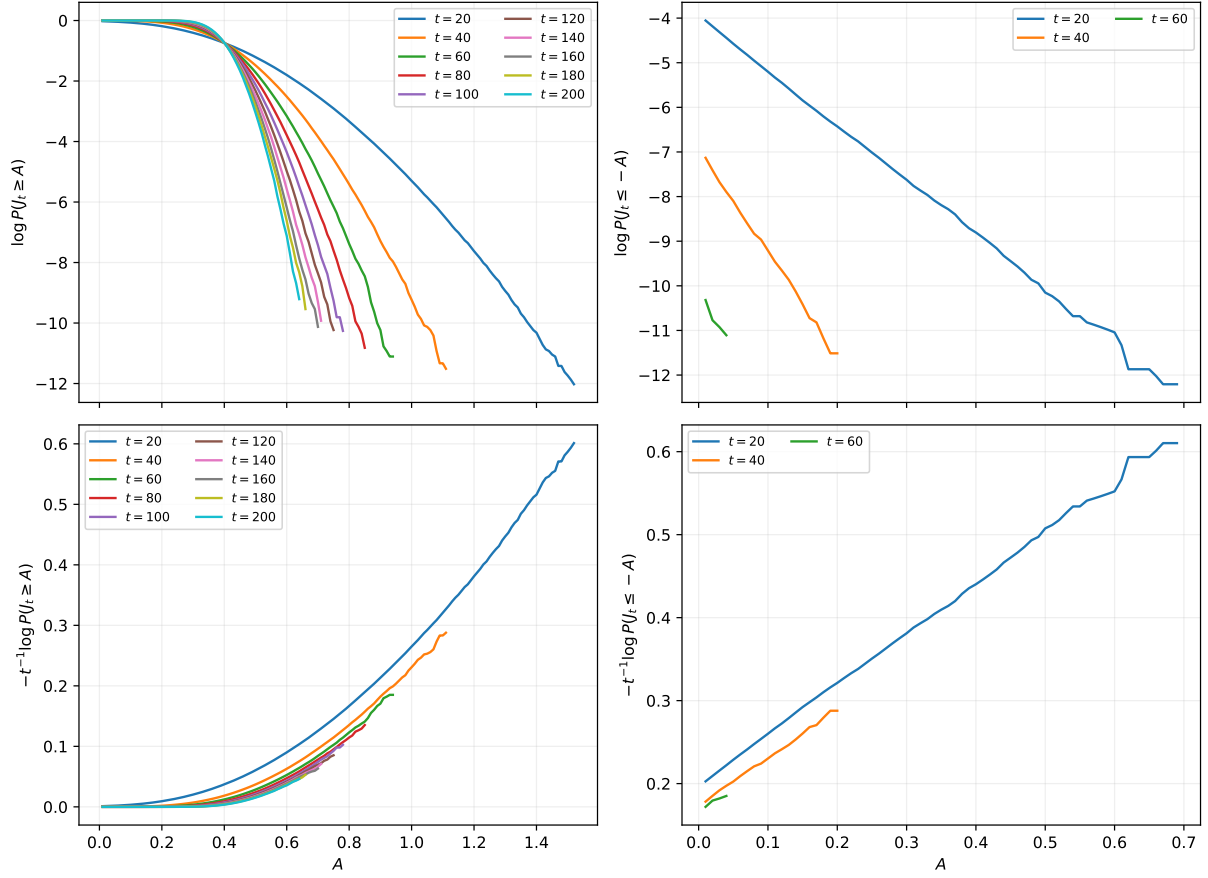


Figure 5: Raw two-tail log probabilities and scaled survival-rate proxies $-t^{-1} \log P$ versus threshold and averaging time for $n = 10$. Collapse of the scaled curves is a large-deviation diagnostic, not by itself an FT test.

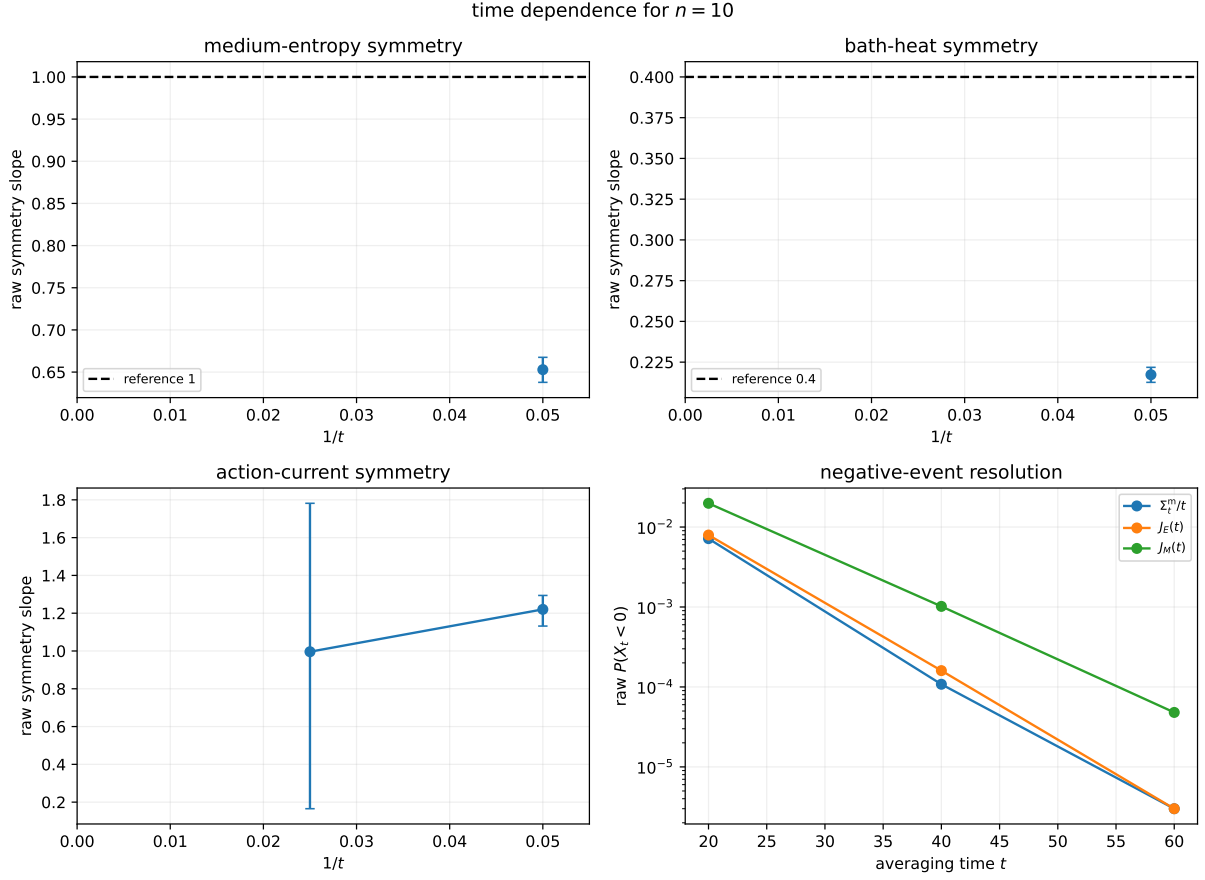


Figure 6: Symmetry slopes and raw negative-event resolution versus averaging time for $n = 10$. Missing points have insufficient two-sided raw support.

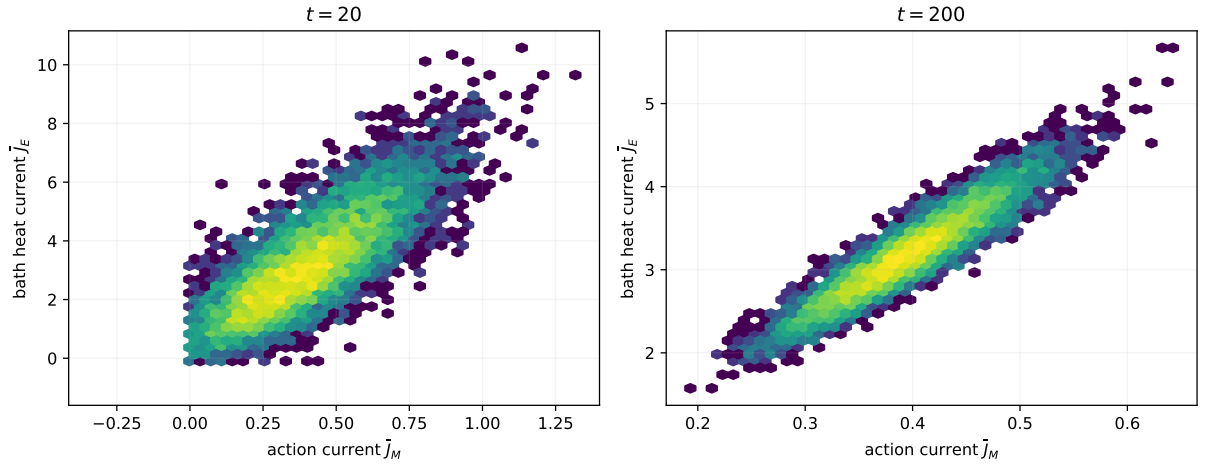


Figure 7: Joint bath-heat/action-current sample for $n = 10$.

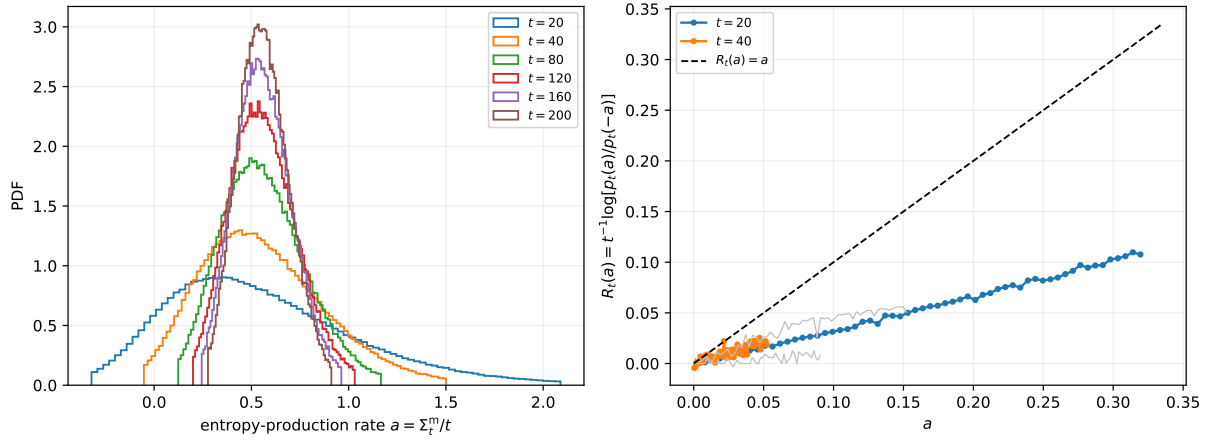


Figure 8: Medium-entropy PDF and raw/plus-four symmetry diagnostic for $n = 20$.

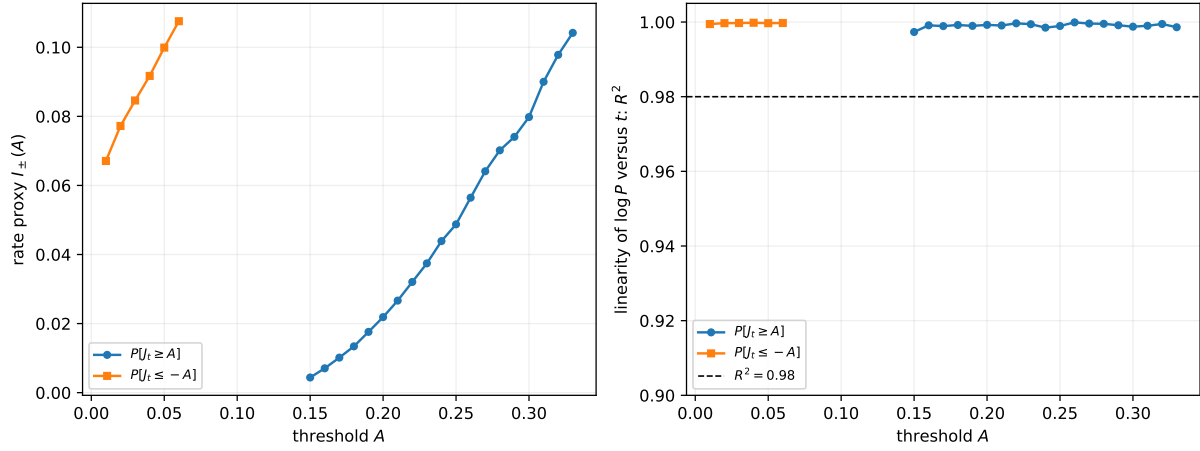


Figure 9: Descriptive action-tail rate proxies and $\log P$ -versus- t linearity for $n = 20$. Only raw-count-qualified rare-tail windows are shown.

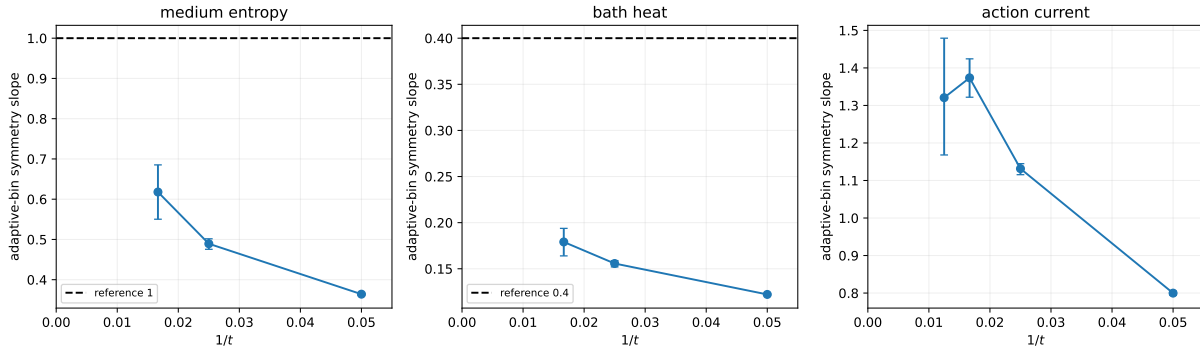


Figure 10: Adaptive-range symmetry-slope robustness check for $n = 20$. The horizontal references apply only to medium entropy and bath heat.

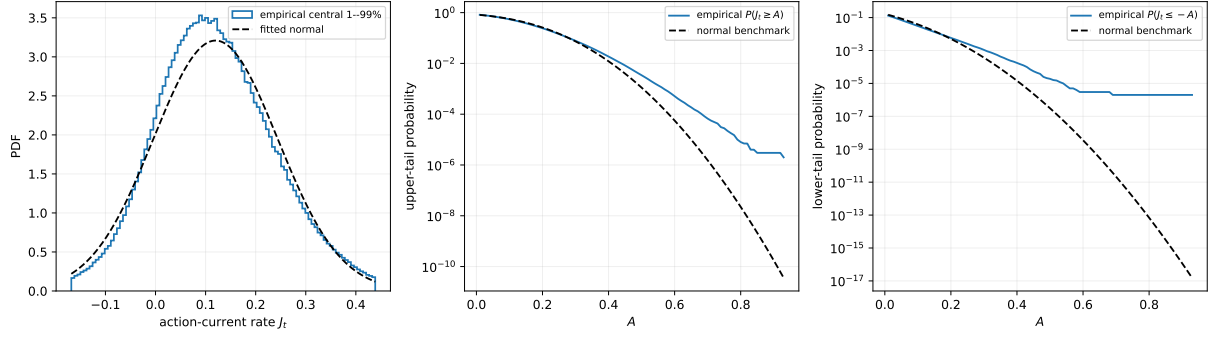


Figure 11: Central action-current histogram and both normal-tail benchmarks for $n = 20$ at $t = 20$.

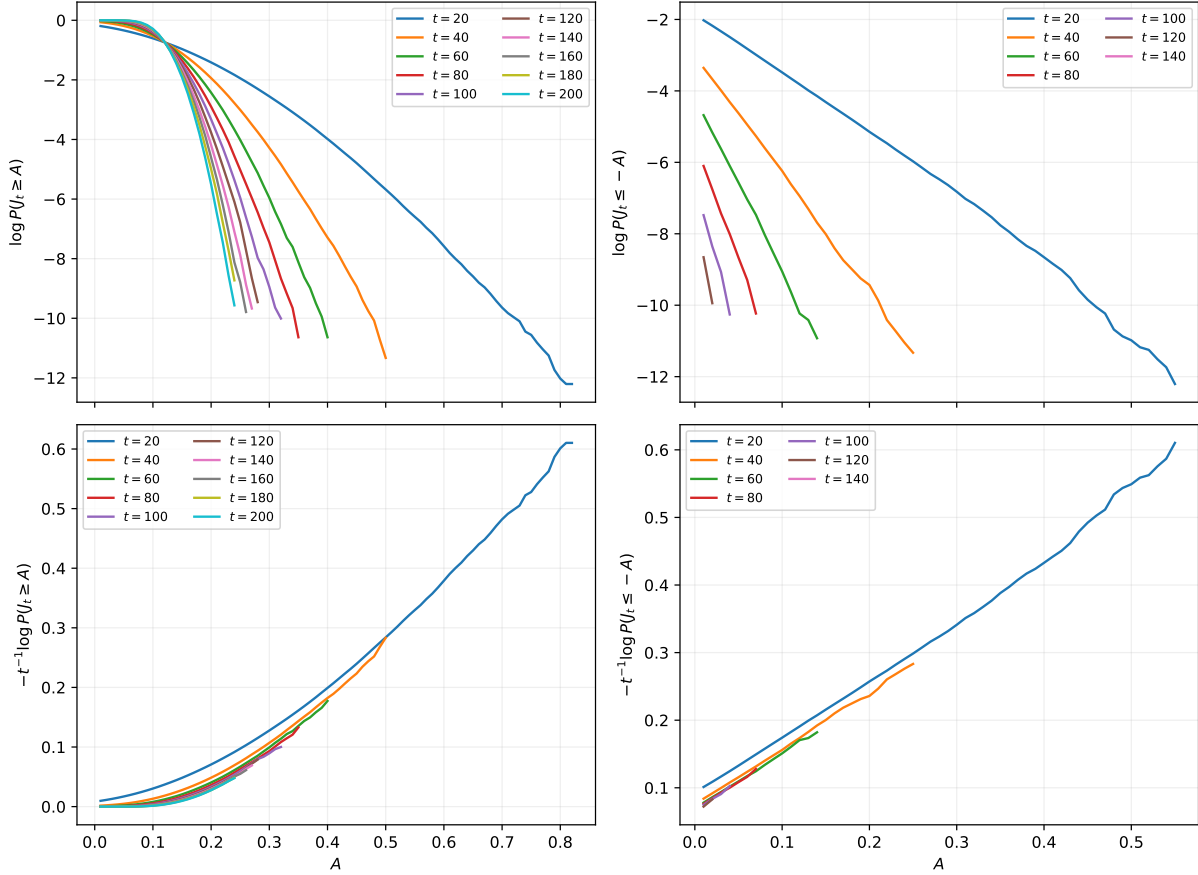


Figure 12: Raw two-tail log probabilities and scaled survival-rate proxies $-t^{-1} \log P$ versus threshold and averaging time for $n = 20$. Collapse of the scaled curves is a large-deviation diagnostic, not by itself an FT test.

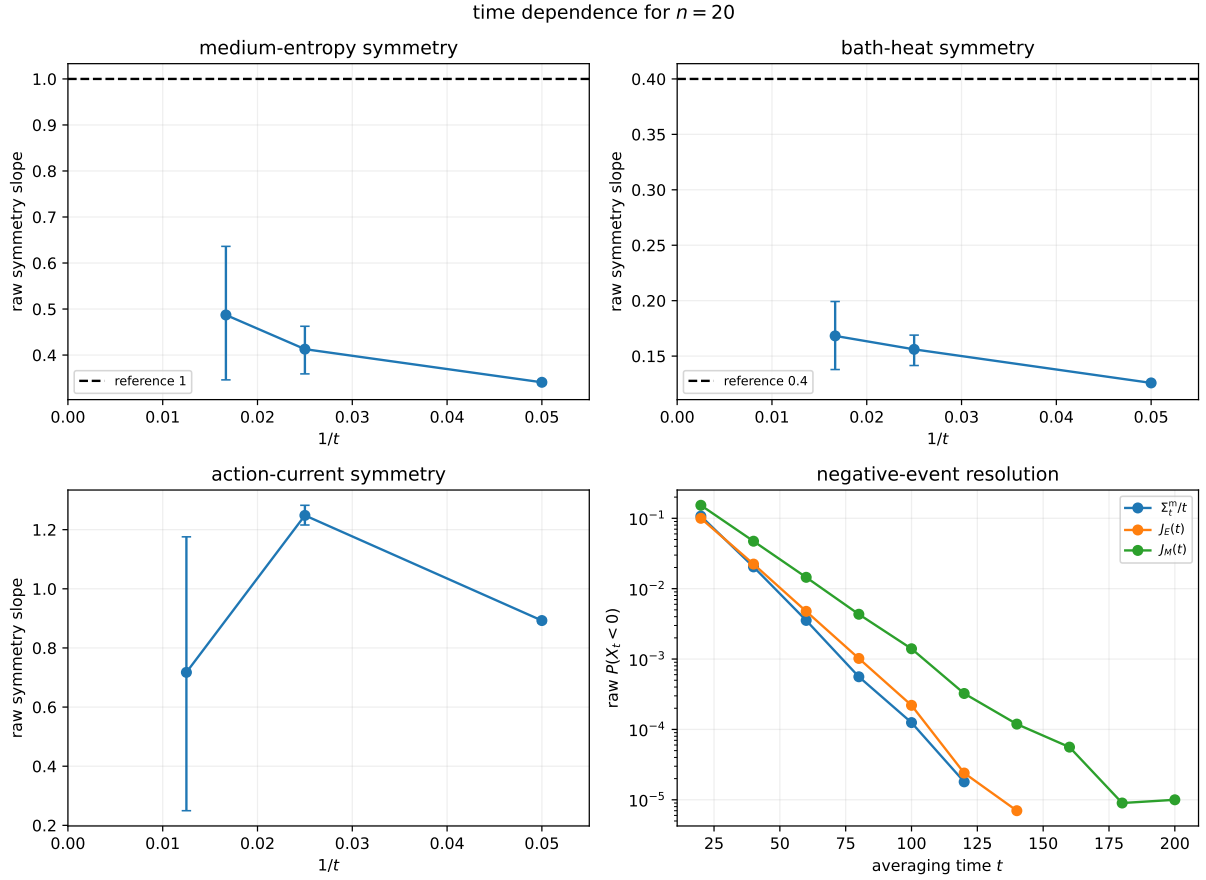


Figure 13: Symmetry slopes and raw negative-event resolution versus averaging time for $n = 20$. Missing points have insufficient two-sided raw support.

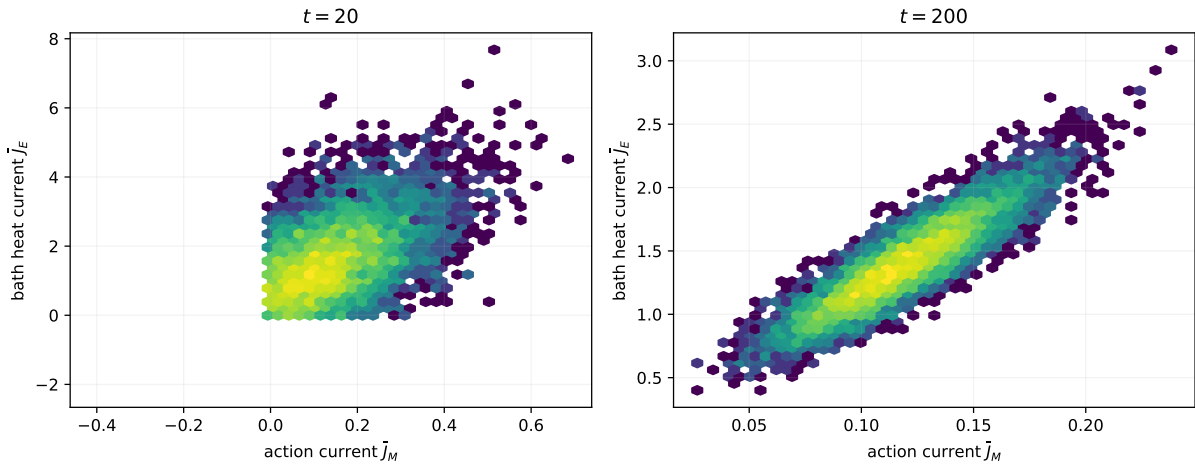


Figure 14: Joint bath-heat/action-current sample for $n = 20$.

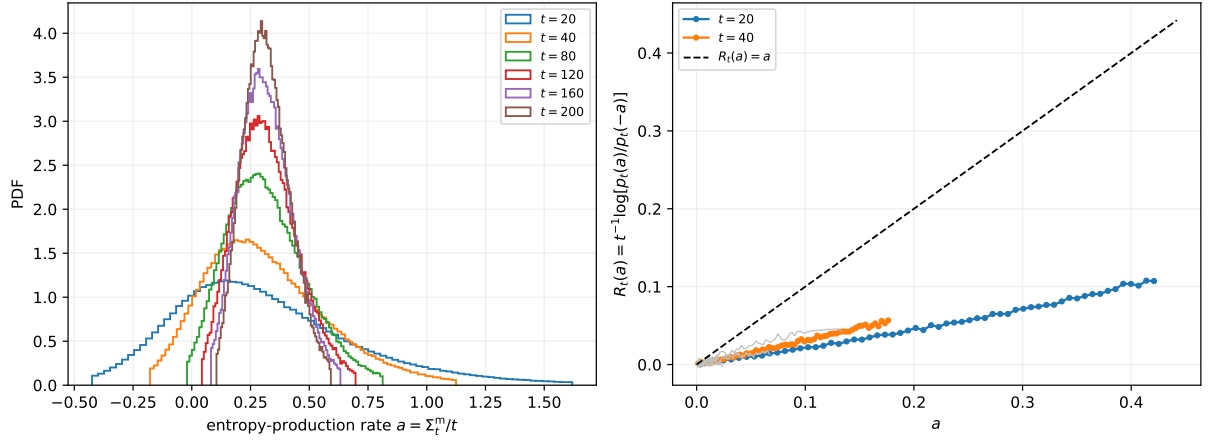


Figure 15: Medium-entropy PDF and raw/plus-four symmetry diagnostic for $n = 30$.

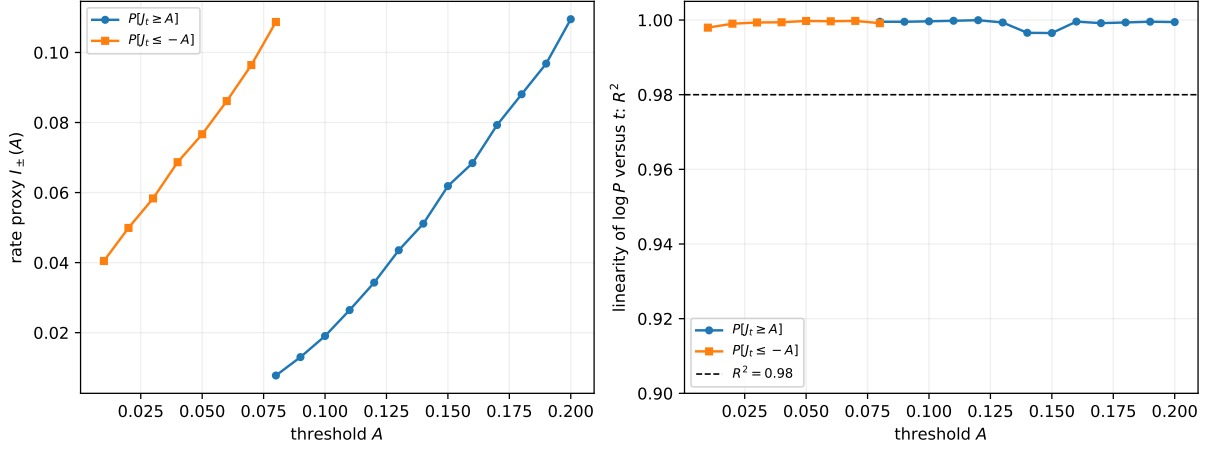


Figure 16: Descriptive action-tail rate proxies and $\log P$ -versus- t linearity for $n = 30$. Only raw-count-qualified rare-tail windows are shown.

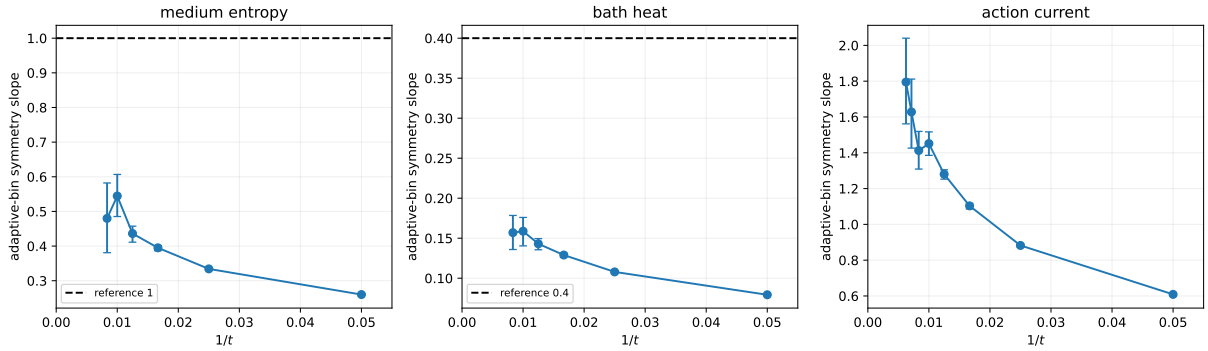


Figure 17: Adaptive-range symmetry-slope robustness check for $n = 30$. The horizontal references apply only to medium entropy and bath heat.

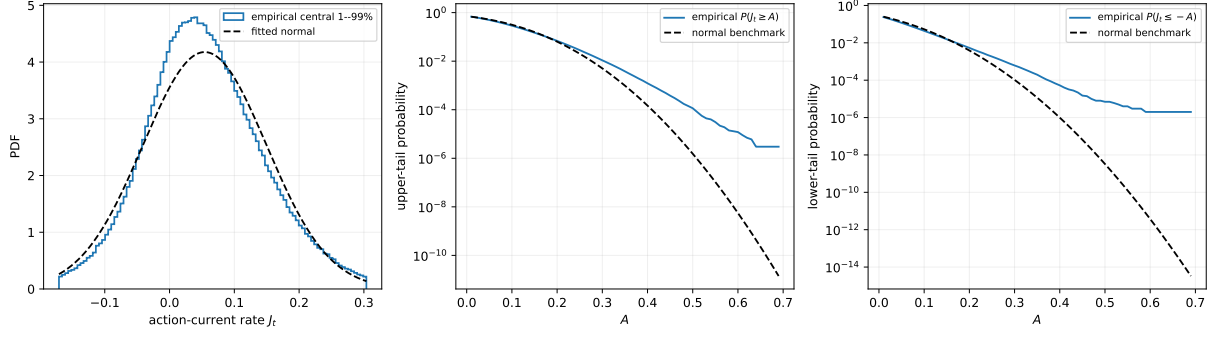


Figure 18: Central action-current histogram and both normal-tail benchmarks for $n = 30$ at $t = 20$.

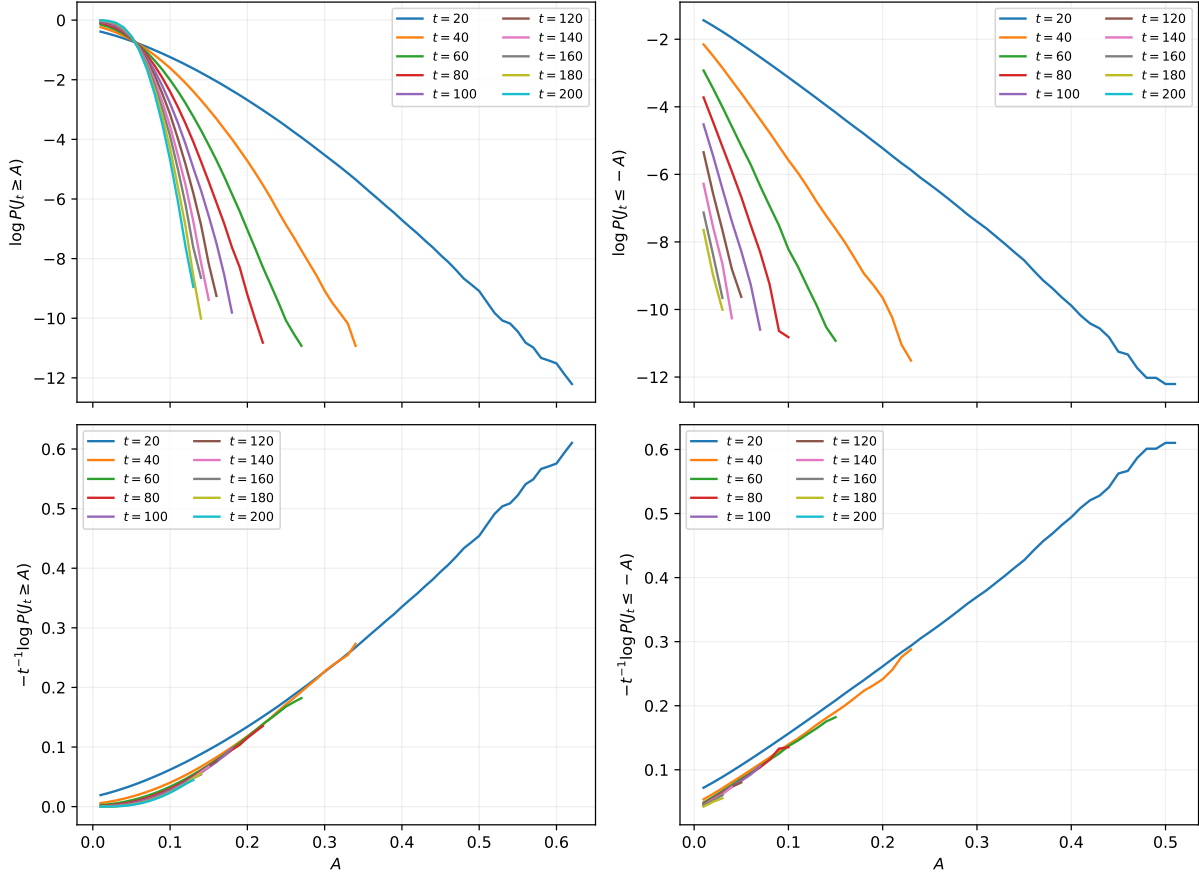


Figure 19: Raw two-tail log probabilities and scaled survival-rate proxies $-t^{-1} \log P$ versus threshold and averaging time for $n = 30$. Collapse of the scaled curves is a large-deviation diagnostic, not by itself an FT test.

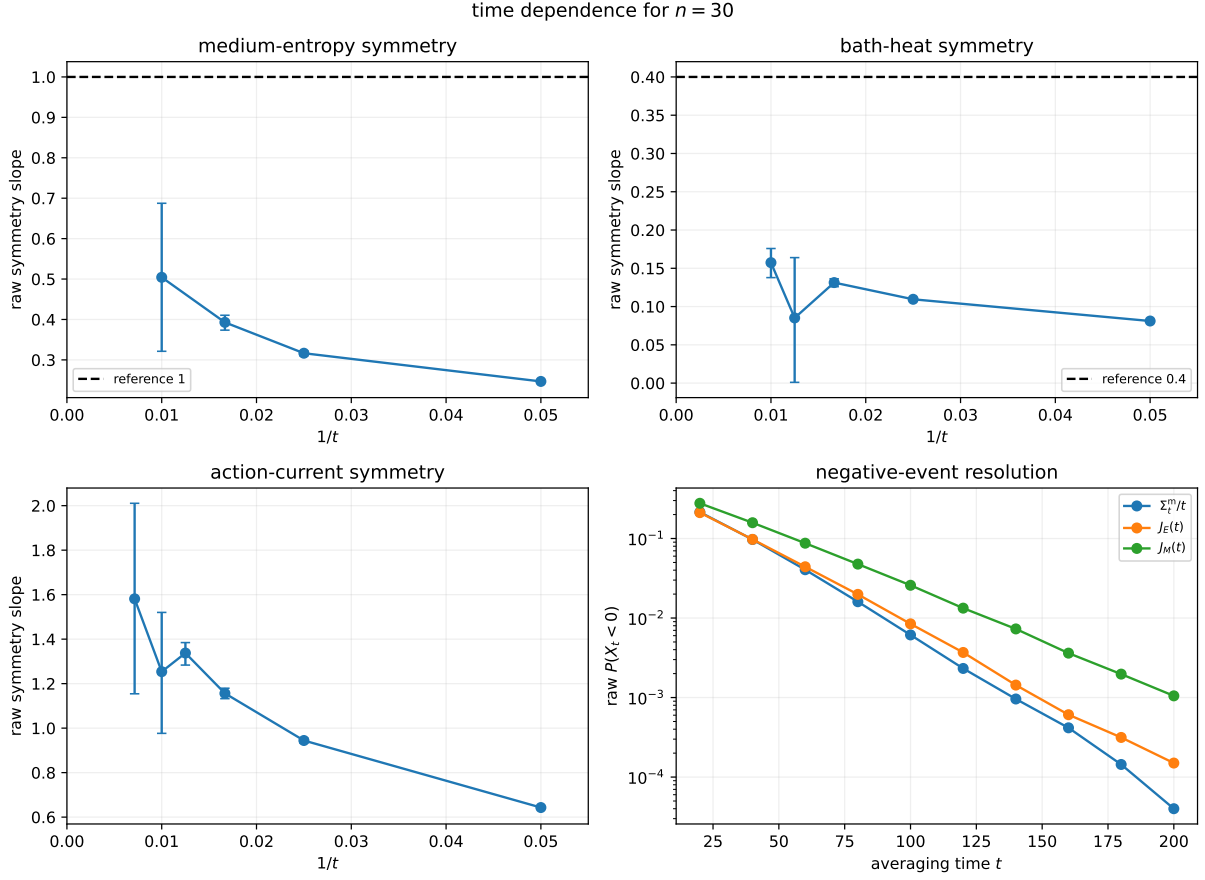


Figure 20: Symmetry slopes and raw negative-event resolution versus averaging time for $n = 30$. Missing points have insufficient two-sided raw support.

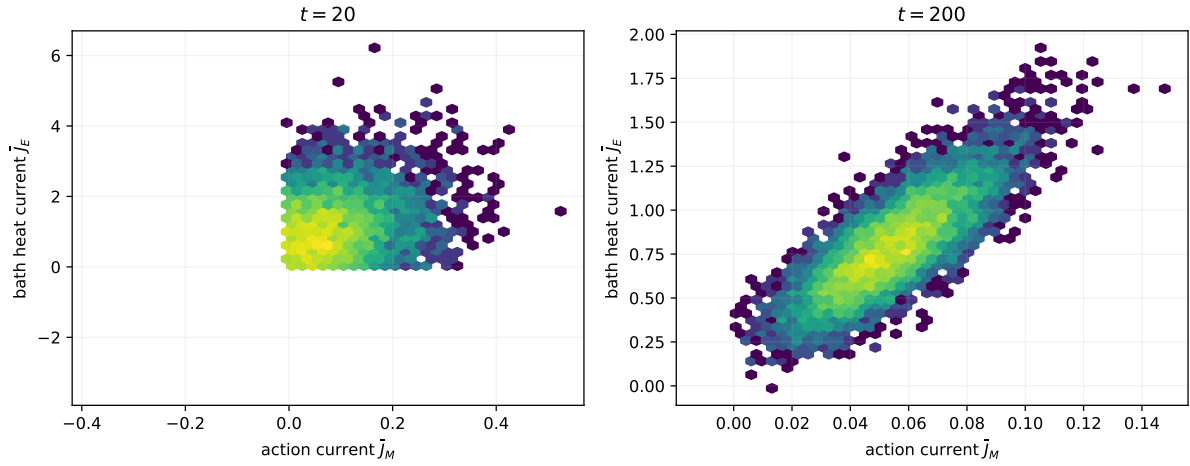


Figure 21: Joint bath-heat/action-current sample for $n = 30$.

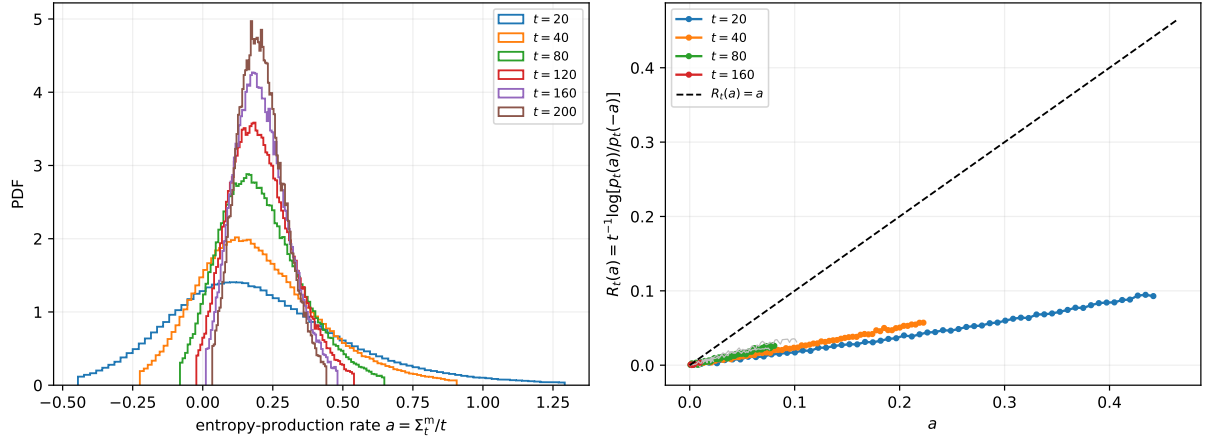


Figure 22: Medium-entropy PDF and raw/plus-four symmetry diagnostic for $n = 40$.

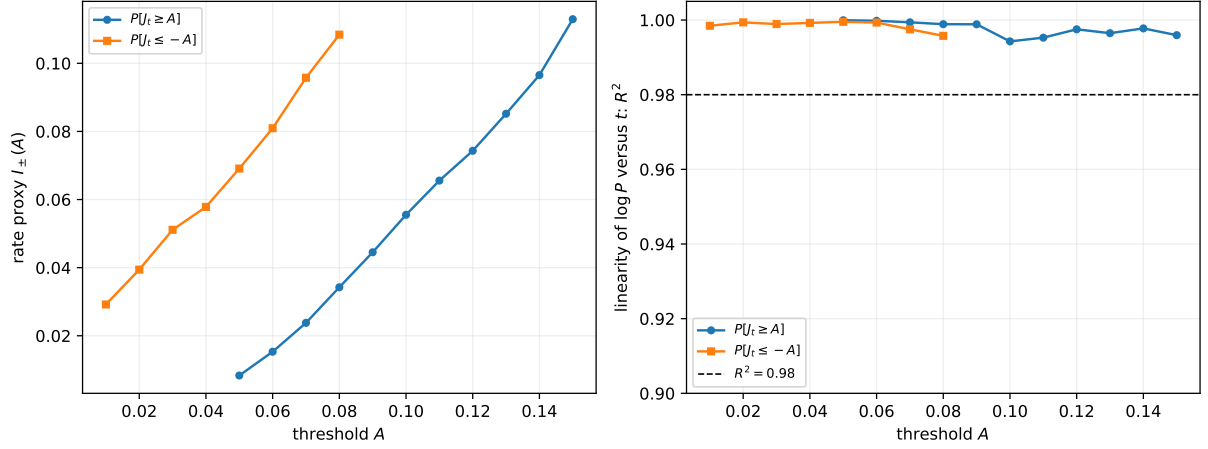


Figure 23: Descriptive action-tail rate proxies and $\log P$ -versus- t linearity for $n = 40$. Only raw-count-qualified rare-tail windows are shown.

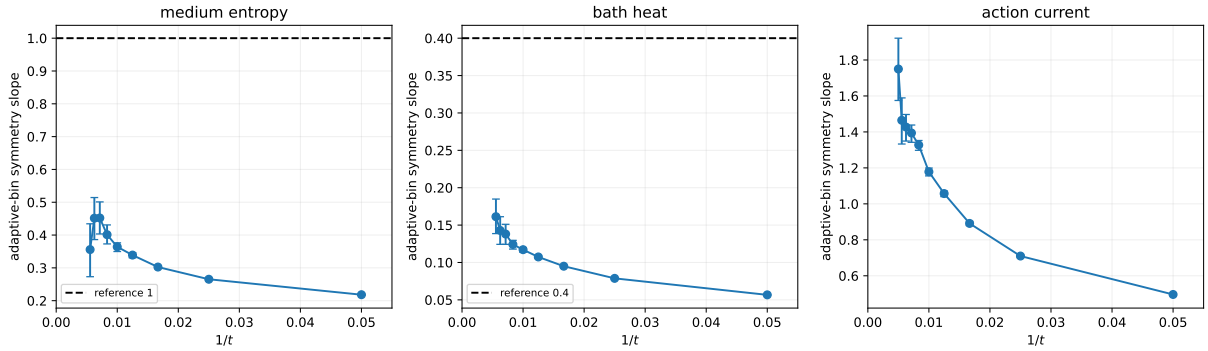


Figure 24: Adaptive-range symmetry-slope robustness check for $n = 40$. The horizontal references apply only to medium entropy and bath heat.

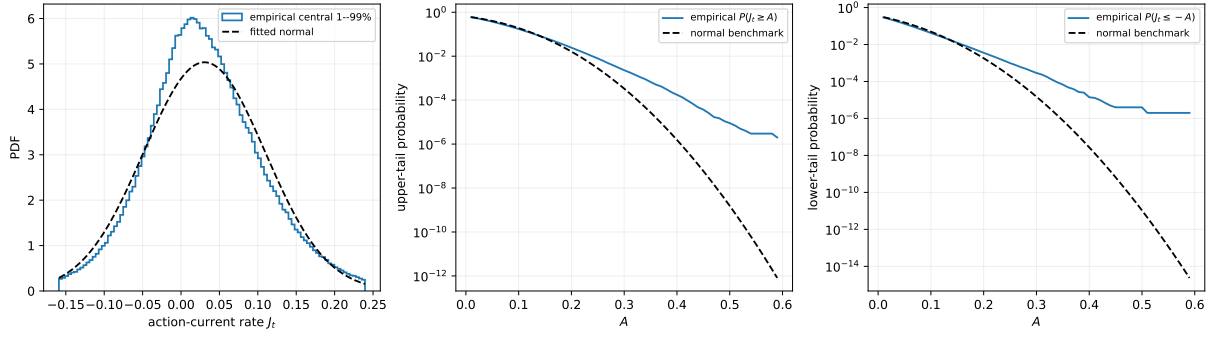


Figure 25: Central action-current histogram and both normal-tail benchmarks for $n = 40$ at $t = 20$.

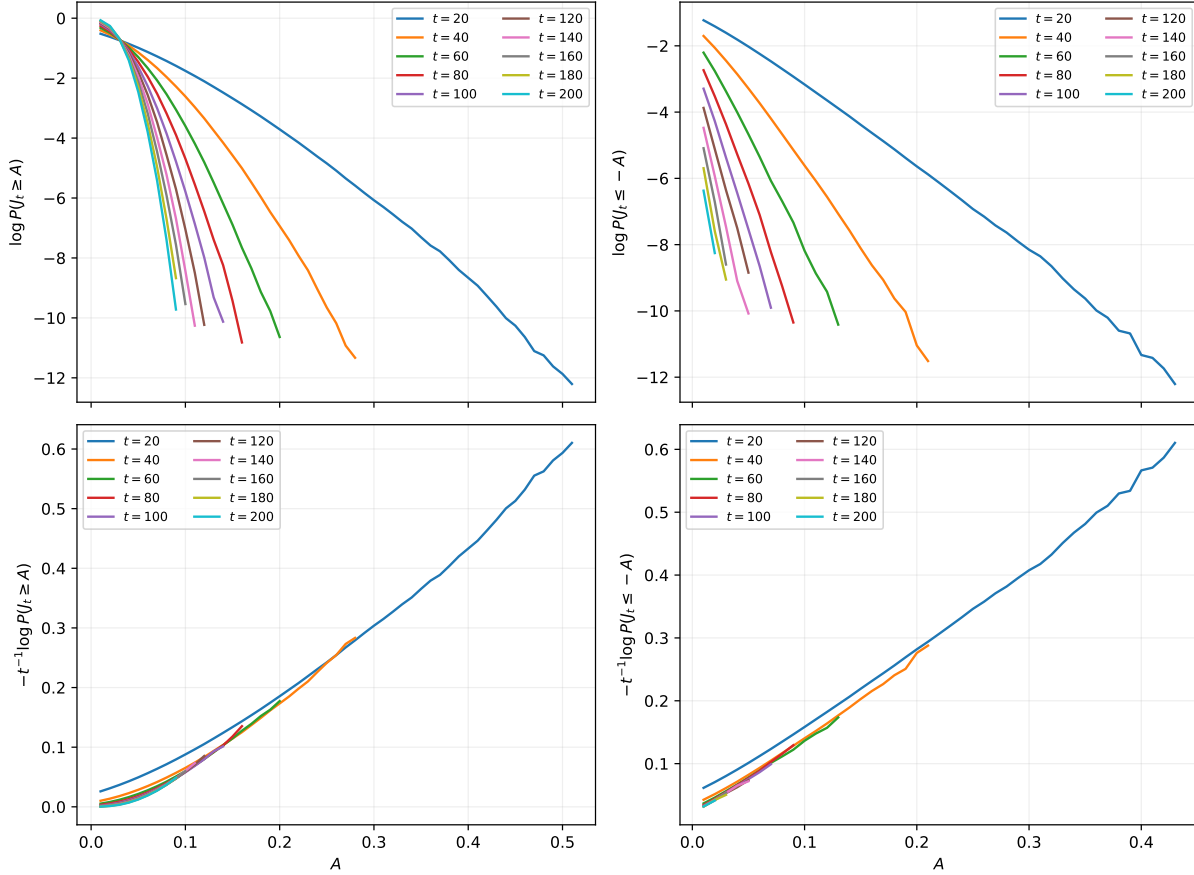


Figure 26: Raw two-tail log probabilities and scaled survival-rate proxies $-t^{-1} \log P$ versus threshold and averaging time for $n = 40$. Collapse of the scaled curves is a large-deviation diagnostic, not by itself an FT test.

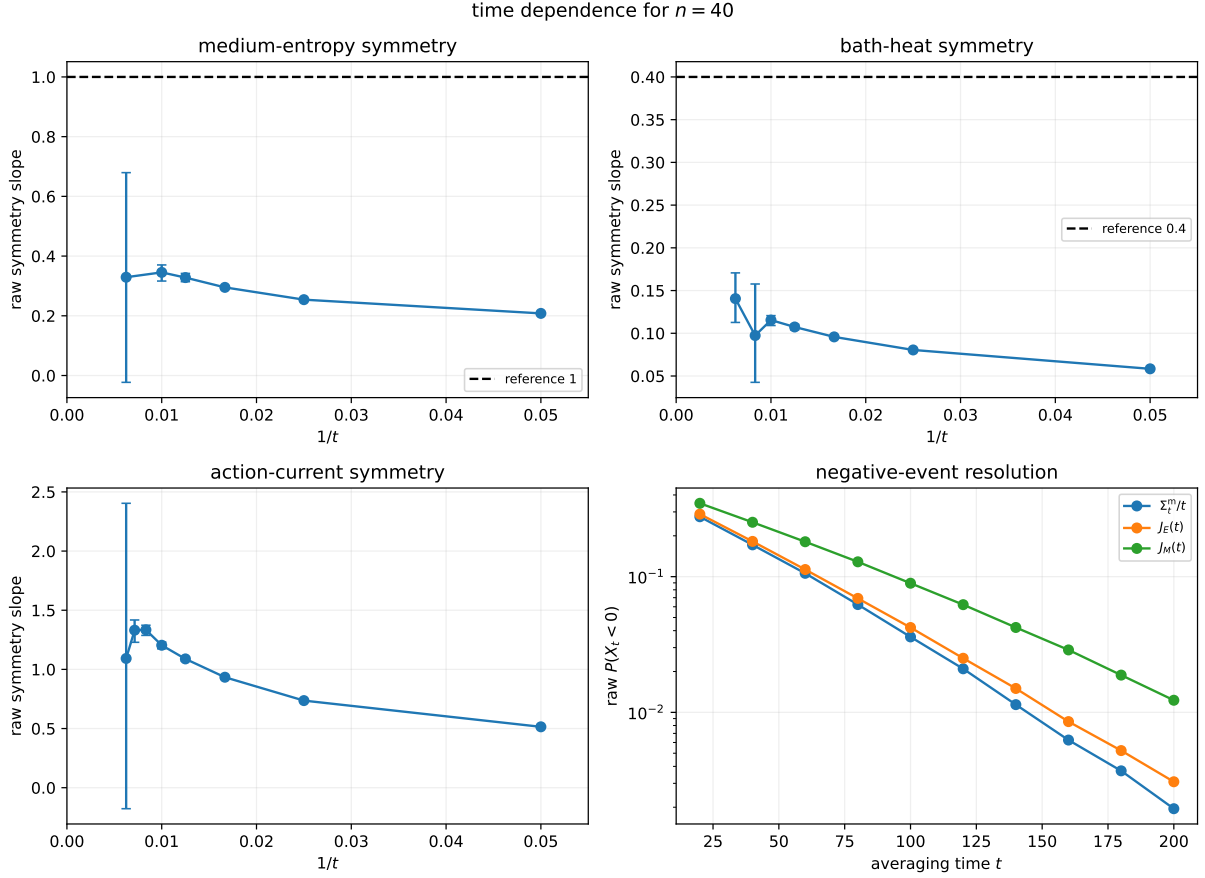


Figure 27: Symmetry slopes and raw negative-event resolution versus averaging time for $n = 40$. Missing points have insufficient two-sided raw support.

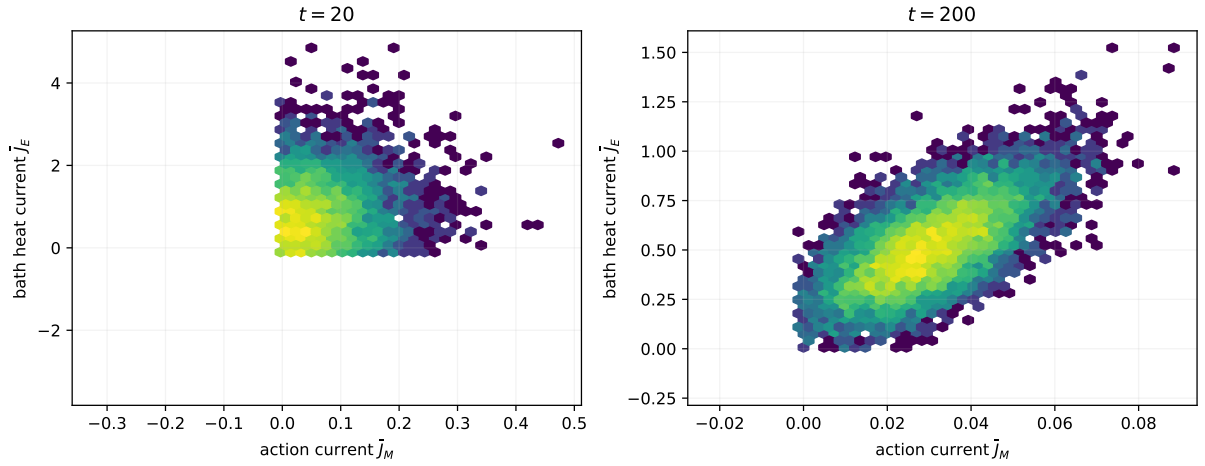


Figure 28: Joint bath-heat/action-current sample for $n = 40$.

Interpretation guardrail

A finite-time detailed FT is a statement about total trajectory entropy, $\Delta s_{\text{tot}} = \Sigma_t^{\text{m}} + \Delta s_{\text{sys}}$. The unknown NESS system-entropy endpoint term is not reconstructed here. Consequently, a deviation of the medium-entropy slope from one is reported as “not verified in the sampled window,” not as a universal violation. Conversely, visual linearity or a Gaussian-looking center is not by itself evidence for the FT.